# Energy transformation

Energy transformation is the process of energy changing from one type of energy to another. This process is happening all the time in the world around us and even in us. When we eat food our body uses the chemical energy in the food and transforms it into mechanical energy (so that we can move), a new form of chemical energy or thermal energy.

Look at this diagram of a torch

The chemical energy in the batteries changes into electrical energy which moves along the wires to the bulb (globe). In the bulb the electrical energy is changed into heat energy and then light energy.

We can show this as an energy flow diagram.

Chemical energy ➔ electrical energy ➔heat energy➔ light energy

Try to write this one out as an energy flow diagram. Theo puts batteries into his portable ipod sation. Electricity runs to the speakers where the electrical energy is changed into sound energy.

_____ energy➔ _____energy➔ _____energy

## Energy transfer

Energy transfer is where energy moves from one place to another.

Throwing a ball involves the transfer of energy from _____ to the ball. The energy that powers your house is _____ from the power station to your home.

## Where does the energy go?

Energy cannot be created or _____.
Energy can be transferred or transformed.